

Digital self-comb transition boundaries — np.diff

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What this is

Eight files bracketing the three genuine on/off transitions of the **digital self-comb** (1.953125 MHz = exactly 8.000 channels, channel-locked, residue 0 mod 8) in the early part of its 07-17/18 era.

#	ON → off	off → ON	extra context
1	corr_20260717_154410Z.h5 (15:44:10)	corr_20260717_154619Z.h5 (15:46:19)	—
2	corr_20260717_155704Z.h5 (15:57:04)	corr_20260717_155912Z.h5 (15:59:12)	—
3	corr_20260717_161623Z.h5 (16:16:23)	corr_20260717_162916Z.h5 (16:29:16)	...162707Z before, ...163125Z after

These are the three transitions attributed to the comb itself. The other 17 box-air boundaries in the era are daemon transitions, a daemon outage, or rfswitch/VNA states where the antenna was not connected — i.e. the instrument stopped looking, not the comb stopping. Those are deliberately excluded.

Transition 3 is bracketed with two extra files because a per-integration check (see "Where transition 3 actually happens") showed the file-level label is misleading there: ...162916Z is an isolated single file — off on both inputs immediately before and immediately after it — and the step inside it is at the very end, not the start. ...162707Z and ...163125Z are its contiguous neighbours (129 s file cadence, verified, no gap).

Purpose: visually confirm and characterise what actually changes at these boundaries. This notebook shows the data; it does not settle the comb's physical origin.

Provenance

- Boundary determination: `marjum-2026-07/flags/diag/comb_inventory.jsonl` (best-phase tooth contrast at 8.000 ch, `lock8`), operating point `lock8 >= 30` with phase 0.
- Data access: `eigsep_data.index.MetadataIndex` + `eigsep_data.bundle.load_bundle` — the canonical loaders. The cross correlation is available through `antenna="cross"` , which resolves the `"04"` key per file and fixes conjugation orientation, so there is no reason to drop to raw `h5py` here.
- Units: raw correlator counts throughout. Frequencies in MHz, times UTC.

numpy 2.2.6

Load

`patterns=` restricts the index to just these six files, and `cache=False` keeps the notebook from writing an `.eigsep_index.h5` sidecar into the shared campaign data directory. Both are deliberate: this is a six-file study and it should leave no trace in shared state.

```
indexed 1920 integrations from 8 files
skipped: none
--- box-gnd (auto, input 0) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-gnd' via keys ['0']
8 files used, 0 skipped
dtype int32

--- box-air (auto, input 4) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'box-air' via keys ['4']
8 files used, 0 skipped
dtype int32

--- cross (box-gnd x box-air) ---
1920 rows x 1024 channels (0.000-249.756 MHz)
antenna 'cross' via keys ['04']
8 files used, 0 skipped
dtype complex128
```

Timebase

Every one of these six files has `sync_consistent = True` , so the index's `time_best` is trustworthy here and is used directly for the UTC labels below.

That is worth stating explicitly because it is not generally true of this campaign: `header/times` is corrupt in 642 of 5,120 files (May-2026 dates, negative spans), and the standing rule is to fall back to the filename close time. These six need no such fallback — checked, not assumed.

	role	n_int	sync_ok	utc_start	utc_end	lock8_box_air	lock8_box_gnd
file							
corr_20260717_154410Z.h5	ON->off	240	True	15:42:03	15:44:11	0.0	0.0
corr_20260717_154619Z.h5	off->ON	240	True	15:44:11	15:46:20	164.6	399.5
corr_20260717_155704Z.h5	ON->off	240	True	15:54:56	15:57:04	19.9	4.4
corr_20260717_155912Z.h5	off->ON	240	True	15:57:05	15:59:13	136.2	362.2
corr_20260717_161623Z.h5	ON->off	240	True	16:14:15	16:16:24	0.0	0.0
corr_20260717_162707Z.h5	before	240	True	16:25:00	16:27:08	0.0	0.0
corr_20260717_162916Z.h5	off->ON	240	True	16:27:08	16:29:17	64.6	212.4
corr_20260717_163125Z.h5	after	240	True	16:29:17	16:31:26	0.0	0.0

all six files: `sync_consistent = True`

int32 accumulator wrap — repaired on the autos only

The correlator's auto accumulators are int32 and **wrap**. Autos are non-negative by construction, so `raw < 0` detects a wrap with no threshold (`eigsep_data.flagging.detectors.overflow_mask`). In this six-file window box-gnd has ~0.22% wrapped samples and box-air ~0.001%.

This matters more here than almost anywhere else: a wrap is a $\pm 2^{32}$ step, and **this notebook differences along time**, so a single unrepaired wrap produces a 4×10^9 -count spike that sets the colour scale and hides everything real.

The cross correlation is deliberately left alone. A cross visibility is genuinely signed in both real and imaginary parts, so `raw < 0` carries no information about wrapping there — applying the auto repair to it would corrupt real data.

```
box-gnd wraps repaired: 4635
box-air wraps repaired: 19
cross left as-is (complex, signed): dtype complex128
1024 channels, 0.000-249.756 MHz
```

Differencing

`np.diff` is taken **within each file**, never across a file join. The six files are not contiguous in time (there are gaps of minutes between them), so differencing a naively concatenated stack would manufacture a bright artificial row at every boundary that has no physical meaning.

Each file's 240 integrations therefore yield 239 difference rows, and a row of `NaN` is inserted between files as a visual separator.

For the cross, the displayed quantity is $\Delta|V|$ — the signed change in visibility amplitude — so that all three panels show the same kind of quantity and can share a colour interpretation. (`|ΔV|` , which also responds to pure phase rotation, is a different question and is not what is plotted here.)

stacked diff shape: (1919, 1024) (8 files x 239 diff rows + 7 separators)
antenna-to-antenna difference rows: 1447 of 1919

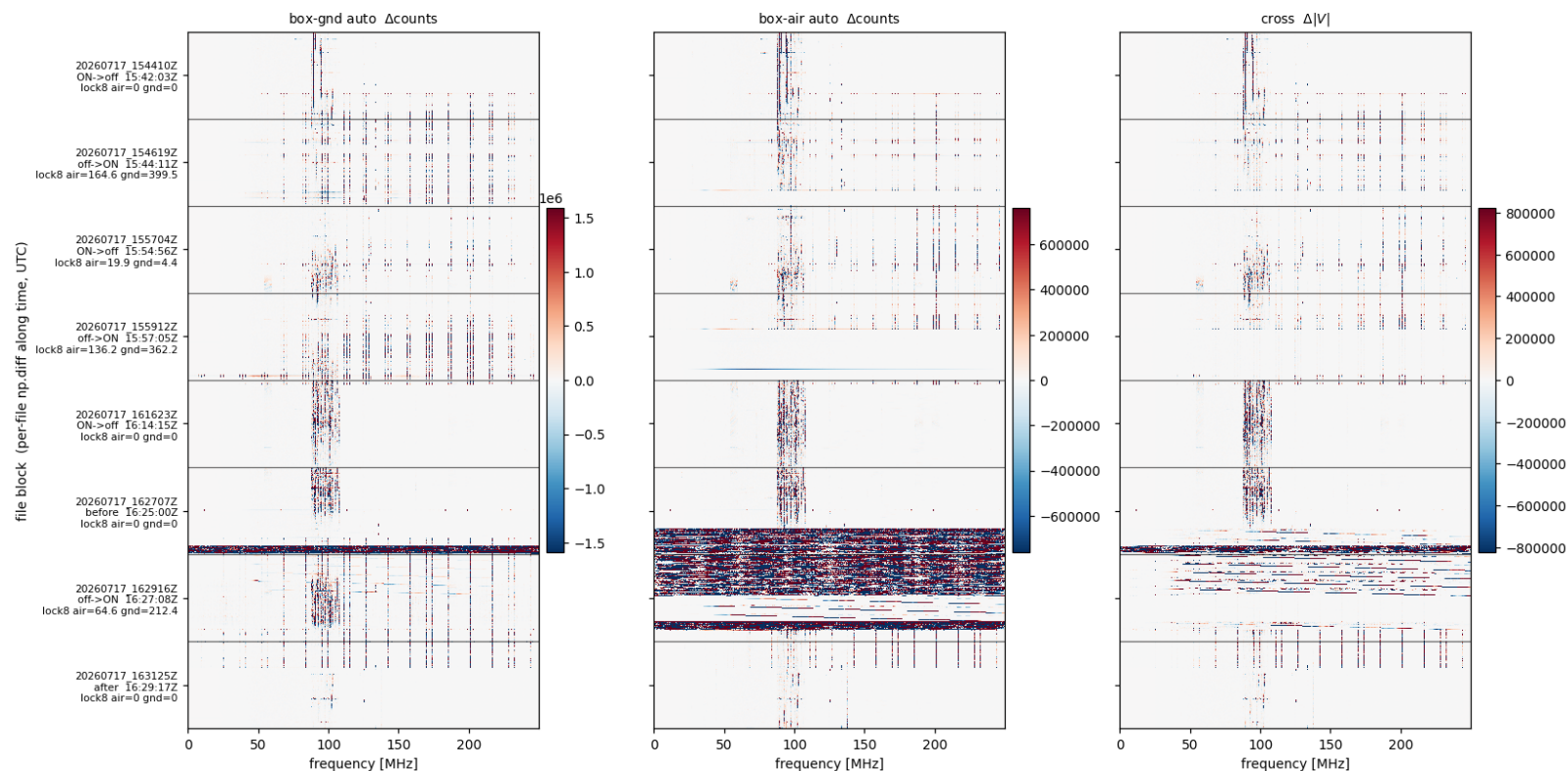
The figure

Three panels, `sharex=True, sharey=True`: box-gnd auto, box-air auto, and the cross amplitude. Frequency runs along x; the concatenated per-file difference rows run down y, with each file's block labelled by name, role and UTC start.

The colour scale is symmetric about zero and set per panel — the three panels differ in absolute scale by orders of magnitude, so a shared colour scale would render two of them blank.

It is set from the **antenna-to-antenna difference rows only**. This is not cosmetic. `corr_20260717_162916Z.h5` runs a VNA sweep, and its switch steps reach $\sim 10^8$ counts — two orders of magnitude above anything on sky. Scaling to all rows lets that one file's calibration saturate the box-air panel and render the other five files blank. Rows outside the antenna state are still drawn, and simply clip.

Digital self-comb transition boundaries — time-difference waterfalls 2026-07-17, three ON/off transition pairs (raw counts)



Which rows are on sky

The same y geometry, showing the receiver state behind every difference row. Green is an antenna-to-antenna difference (a real on-sky step); grey is any row with a load, noise-source or VNA state at either end — those rows are switch steps and carry no sky information.

antenna-to-antenna
difference rows

20260717 154410Z
ON->off 15:42:03Z
lock8 air=0 gnd=0

20260717 154619Z
off->ON 15:44:11Z
lock8 air=164.6 gnd=399.5

20260717 155704Z
ON->off 15:54:56Z
lock8 air=19.9 gnd=4.4

20260717 155912Z
off->ON 15:57:05Z
lock8 air=136.2 gnd=362.2

20260717 161623Z
ON->off 16:14:15Z
lock8 air=0 gnd=0

20260717 162707Z
before 16:25:00Z
lock8 air=0 gnd=0

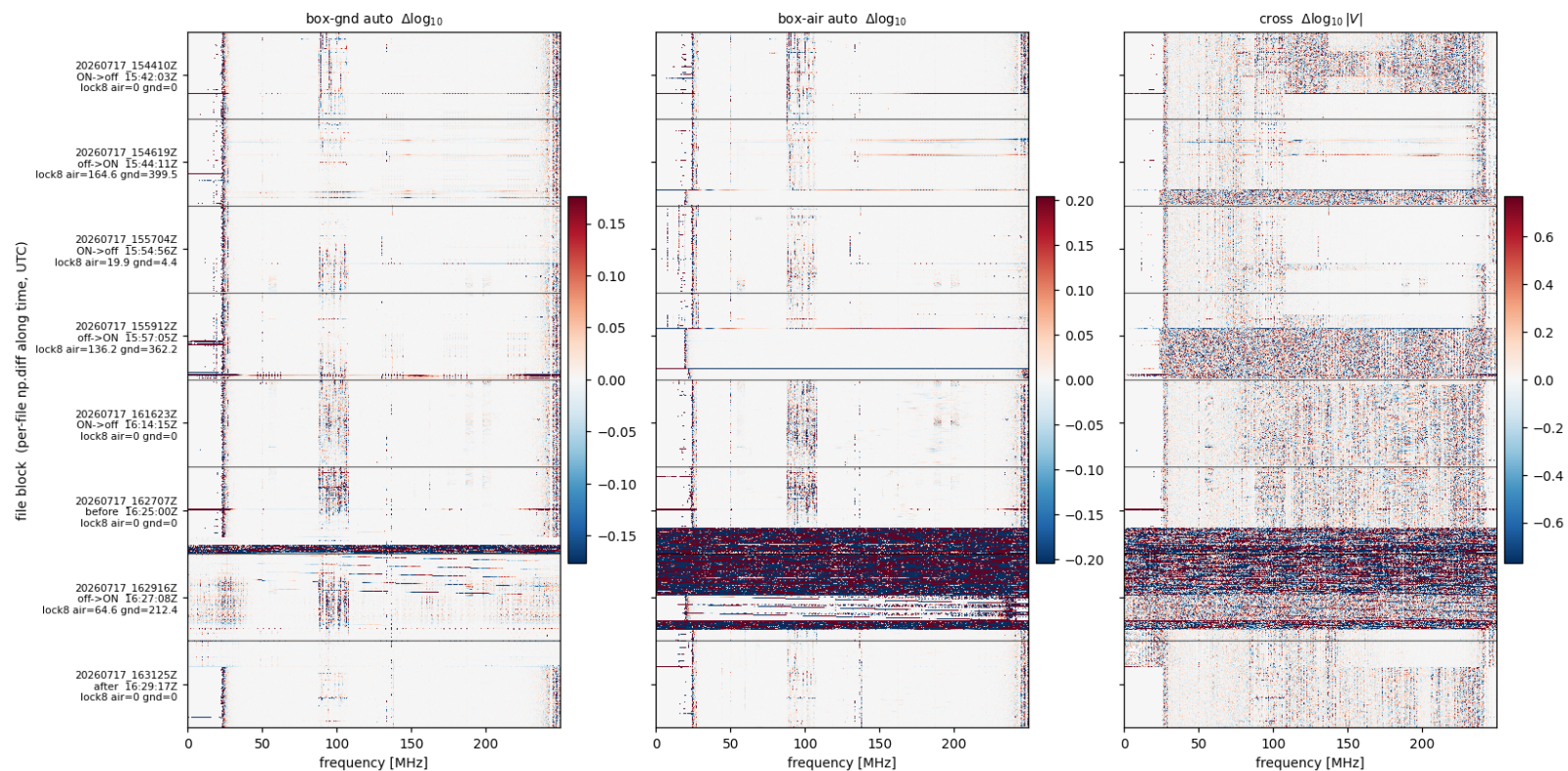
20260717_154410Z: 100.0% on sky
20260717_154619Z: 82.4% on sky
20260717_155704Z: 100.0% on sky
20260717_155912Z: 41.0% on sky
20260717_161623Z: 100.0% on sky
20260717_162707Z: 70.3% on sky
20260717_162916Z: 11.7% on sky
20260717_163125Z: 100.0% on sky

Same difference, log space

The linear panels above are dominated by the brightest channels, which is what a difference of raw counts does. The comb is a narrowband, low-amplitude feature, so the same difference taken in $\log_{10}$ space — i.e. a fractional change per integration — is what actually makes a comb switching on or off legible.

This is the same operation on the same data, not a different selection. The autos use $\log_{10}(\max(\text{counts}, 1))$; the cross uses $\log_{10}(\max(|V|, 1))$.

Same boundaries, difference taken in log space
(fractional change per integration — narrowband features legible)



Where transition 3 actually happens

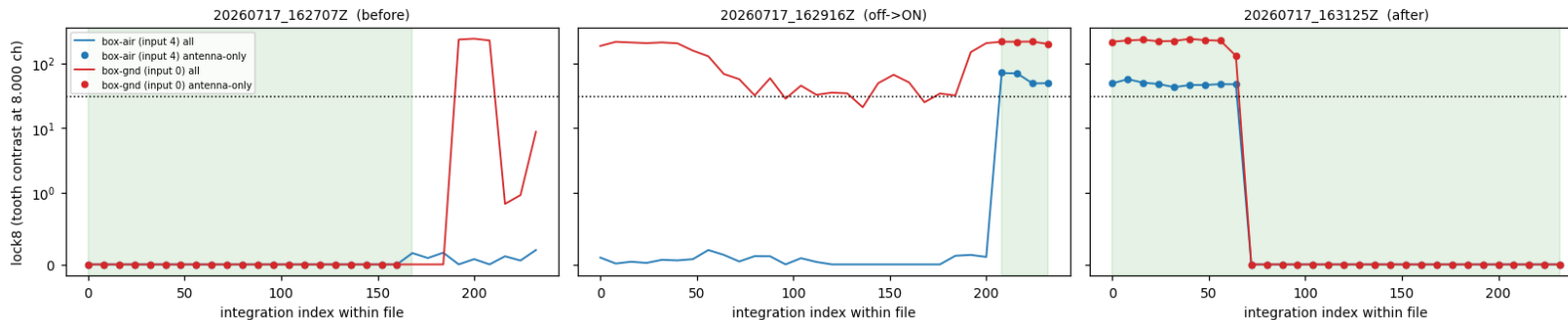
The file-level label says the comb returns at `corr_20260717_162916Z.h5` (16:29:16). A file is 240 integrations over ~129 s, so that label localises the step no better than to the file. The question worth asking directly is where inside the file it happens — and the answer decides whether the notebook is even showing the transition.

Below, `lock8` is recomputed per 8-integration block using `comb_inventory.py`'s own `tooth_contrast_locked` — imported from the script, not reimplemented, so the statistic cannot silently drift from the one that produced the published boundary list. Two traces per input:

- **all** — every integration in the block, whatever the receiver state.
- **antenna-only** — restricted to `RFANT` integrations, which is what `comb_inventory.process()` itself does.

tooth statistic imported from `/mnt/data02/eigsep/data-analysis/scripts/marjum-2026-07/flagging/comb_inventory.py`
traces computed

Transition 3, per-integration: where the comb actually appears
dotted line = `lock8 = 30` operating point; green shading = antenna (RFANT) integrations



20260717_162707Z	RFANT ints 169/240	box-air lock8 max	0.2	box-gnd lock8 max	240.3
20260717_162916Z	RFANT ints 29/240	box-air lock8 max	70.6	box-gnd lock8 max	216.7
20260717_163125Z	RFANT ints 240/240	box-air lock8 max	56.8	box-gnd lock8 max	238.8

What the trace shows — the file-level label is wrong three ways

1. On box-air the step is at the very end of ...162916Z , not the front. lock8 sits at ~0.1 for integrations 0–207 and only rises at **integration 208 of 240 (87% into the file, 16:29:00)**, reaching 49–72 for the final 32. Those final integrations are exactly the file's only RFANT integrations. So what box-air records is not the comb switching on — it is the **antenna being reconnected** at the end of a VNA sweep, revealing a comb box-air could not see while the antenna was disconnected. The file-level lock8 = 64.6 is computed entirely from those 29 integrations.

2. On box-gnd the comb turns on in the previous file. ...162707Z box-gnd is flat at 0.0 for integrations 0–191, then jumps to 233/240/225 at integrations 192–215 — onset **16:26:43**, some 2.5 minutes before the file-level boundary. Box-gnd is conducted and does not need the antenna connected, so it can date the onset and box-air cannot.

3. It turns off in the following file, and the file-level statistic misses that entirely. ...163125Z is published as lock8 = 0.0 on both inputs. Per block it is nothing of the kind: the comb is **present for integrations 0–71 on both inputs** (box-gnd 130–239, box-air 43–57), then stops hard at **integration 72, 16:29:56**, and stays off for the remaining 168. The published 0.0 is a **median artefact** — lock8 is computed from a median over all 240 integrations, so a feature with ~30% duty cycle within a file washes out completely and the file reads as clean.

Putting the three together, transition 3 is not a single-file event at 16:29:16 at all. The actual episode is

16:26:43 → 16:29:56 UTC, about 3 min 13 s, spanning three files.

The boundary list compresses that to one file and dates its start 2.5 minutes late. This is a **systematic** property, not a one-off: any comb episode shorter than roughly half a file is liable to be missed outright, and any episode whose start falls in a cal-heavy stretch will be dated to whenever the antenna next comes back. Short episodes elsewhere in the era should be assumed to be under-counted until checked per-integration.

Adding the two neighbouring files is what makes all of this visible; with the original six it was invisible by construction.

Read this before interpreting the panels: the rfswitch confound

The receiver switches between the antenna (`RFANT`), an ambient load (`RFAMB`), a noise source (`RFN0N`) and several VNA states on a calibration cadence within these files. Differencing along time responds to every one of those switches, and a switch is a far larger step than the comb.

In this particular six-file selection the cadence is **not** balanced across the two roles:

	state file	role	lock8	RFAMB	RFANT	RFNON	UNKNOWN	VNAAMB	VNAANT	VNAL	VNANOFF	VNANON	VNAO	VNAR
corr_20260717_154410Z.h5	ON->off		0.0	0	240	0	0	0	0	0	0	0	0	
corr_20260717_154619Z.h5	off->ON		164.6	0	198	40	2	0	0	0	0	0	0	
corr_20260717_155704Z.h5	ON->off		19.9	0	240	0	0	0	0	0	0	0	0	
corr_20260717_155912Z.h5	off->ON		136.2	27	99	110	4	0	0	0	0	0	0	
corr_20260717_161623Z.h5	ON->off		0.0	0	240	0	0	0	0	0	0	0	0	
corr_20260717_162707Z.h5	before		0.0	0	169	0	7	0	22	21	0	0	0	
corr_20260717_162916Z.h5	off->ON		64.6	0	29	0	12	22	0	22	22	22	22	22
corr_20260717_163125Z.h5	after		0.0	0	240	0	0	0	0	0	0	0	0	

pure-RFANT (no cal switching in the file):

corr_20260717_154410Z.h5	role=ON->off	pure_RFANT=True
corr_20260717_154619Z.h5	role=off->ON	pure_RFANT=False
corr_20260717_155704Z.h5	role=ON->off	pure_RFANT=True
corr_20260717_155912Z.h5	role=off->ON	pure_RFANT=False
corr_20260717_161623Z.h5	role=ON->off	pure_RFANT=True
corr_20260717_162707Z.h5	role=before	pure_RFANT=False
corr_20260717_162916Z.h5	role=off->ON	pure_RFANT=False
corr_20260717_163125Z.h5	role=after	pure_RFANT=True

Every ON -> off **file is 100% RFANT ; every** off -> ON **file contains calibration or VNA states.** In the original six-file selection the split was perfect, 3/3 and 3/3.

That is a confound, and it has to be stated plainly: within those six files alone, "comb detected" and "calibration cadence running" are not separable, so nothing in the waterfalls above can by itself distinguish the comb switching on from the calibration hardware switching on.

The two files added for transition 3 partly break that degeneracy, which is the main reason they earn their place beyond bracketing:

- ...163125Z is **240/240 RFANT , 100% on sky** and shows the comb present on the antenna for its first 72 integrations and absent for the remaining 168 — an on and an off, both with the antenna connected throughout and no switch anywhere near either. That is the control the original six lacked entirely, and it rules out the reading that the comb is merely "whatever the VNA sweep puts into the band".
- ...162707Z is 70.3% on sky and carries the box-gnd onset at integration ~192, i.e. the comb starts while the antenna is still connected, not at a switch edge.

So for transition 3 the confound is now **broken**, not merely testable: the episode both starts and ends inside antenna-connected data. It remains total for transitions 1 and 2.

It is in any case a property of this small selection and **not** of the era. Across the full 07-17 15:37 → 07-18 03:00 era on box-air, at the same lock8 >= 30 operating point, the association runs the other way:

	pure RFANT	cal/VNA active
comb ON	176	29
comb off	49	56

So era-wide the comb is predominantly detected in clean antenna files, and the six boundary files here are an unrepresentative stretch. The honest reading is that these six make a good visual case study of what a boundary looks like, and a **bad** basis for attribution.

One further caveat specific to `corr_20260717_162916Z.h5`: only 29 of its 240 integrations are `RFANT`, so its `lock8 = 64.6` detection rests on 29 antenna integrations, not 240.

Decision requested

1. Does the characterisation above match what you expect a genuine comb on/off boundary to look like, or do the panels instead read as calibration switching to you?
2. **The boundary list needs re-deriving per-integration, and this is now the biggest open item.**
Transition 3 should read **16:26:43 → 16:29:56** across three files, not a single-file event at 16:29:16. Two distinct failure modes are demonstrated above: the file-level `lock8` median **misses** episodes with less than roughly half-file duty cycle (`...163125Z` publishes 0.0 while carrying the comb for 72 integrations), and where the antenna is disconnected it dates the antenna returning rather than the comb starting. Both bias the era's episode count and durations. I have not edited the boundary list. Re-deriving it per-integration on **box-gnd** — the input that sees through a cal cycle — is the fix, and it is a new milestone.
3. If the rfswitch confound is the dominant concern for transitions 1 and 2, the natural next step is to re-pick boundary files **matched on cal state** (both sides pure- `RFANT`) and repeat this figure. Era-wide there are 49 pure- `RFANT` comb-off files to draw from. That is a new selection and a new milestone, so I have not done it here.
4. `lock8 >= 30` is the operating point used to call these transitions; it is defensible (a stable plateau, and 216/220 agreement with beam-analyst's independent detector) but it is a choice, and transition counts move with it.

Not claimed here: anything about the comb's physical origin. This notebook shows morphology at six boundaries; it runs no pointing test, no null control, and no far-field discriminator.